

Mohammad Ful Hossain Seikh

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Professional Objective

To develop expertise in experimental Astroparticle Physics through cutting-edge research at world-class laboratories, while contributing to the advancement of neutrino detection technologies.

Education

Degree	Years	Institution	Performance
Ph.D. in Physics	2020-Present	University of Kansas, USA	3.98/4.0
M.S. in Electrical Engineering	2024-Present	University of Kansas, USA	4.0/4.0
M.Sc. in Physics	2018-2020	Aligarh Muslim University, India	9.170/10 (Gold Medalist)
B.Sc. Honors in Physics	2015-2018	Aligarh Muslim University, India	9.186/10 (Gold Medalist)
Higher Secondary (12th)	2014	CBSE, India	9.120/10 (First Position)
Secondary (10th)	2012	CBSE, India	10/10 (Certificate of Merit)

Research Experience

- **Primary Analyzer, Askaryan Radio Array (ARA) Station 1** 2021-Present
Conducting data analysis of ARA Station 1 for ultra-high energy (UHE) neutrino detection under Prof. D. Besson at University of Kansas (KU).
- **IceCube Neutrino Observatory Monitoring Team** 2021-2025
Served as KU representative for five monitoring shifts, presenting weekly reports to the collaboration.
- **Master's Thesis Research** 2019-2020
Investigated light sterile neutrino oscillations under Prof. M.S. Athar at Aligarh Muslim University (AMU), producing review report "Light Sterile Neutrinos: Status and Prospects" (DOI:10.13140/RG.2.2.13777.66405).
- **Summer Research Program** May-Jul 2019
At Institute of Physics, Bhubaneswar, studied the role of θ_{13} mixing angle in three-flavor neutrino oscillations under Prof. S.K. Agarwalla, producing technical report "Importance of 1-3 Mixing Angle in Three-Flavor Neutrino Oscillations Paradigm" (DOI: 10.13140/RG.2.2.34421.12009).
- **Winter Internship** Dec 2019-Jan 2020
Researched weak interaction physics at IIT-BHU under Dr. G. Abbas.
- **Undergraduate Research Project** 2017-2018
"Quarks & Leptons": detailed study of Standard Model interactions under Prof. M.S. Athar at AMU.

Publications

First-Author Works

- **Seikh, M.F.H.** (2024). "Physics of radio antennas". *European Physics Journal-Special Topics (EPJ-ST)*. arXiv:2411.07507
- **Seikh, M.F.H. for the ARA Collaboration** (2024). "Askaryan Radio Array: searching for the highest energy neutrinos". *EPJ-ST*. arXiv:2411.01761
- **Seikh, M.F.H. & Besson, D.** (2024). "IceCube Neutrino Observatory: A Cold Discovery Hub". *Physics Bulletin*, Physics Department, AMU.

- **Seikh, M.F.H. for the ARA Collaboration** (2023). “Calibration and Physics with ARA Station 1”. *PoS(ICRC2023)1163*. arXiv:2308.07292
- **Seikh, M.F.H.** (2020). “Neutrinos and the 40 Future Routes”. *Physics Bulletin*, AMU. DOI: 10.13140/RG.2.2.27981.10729

Collaborative Works

Full publication record available on:

- Inspire HEP
- Google Scholar
- ResearchGate

Includes contributions to ARA, RNO-G, ARIANNA, IceCube, TAROGE-M and RET collaborations.

Professional Service

Editorial/Review Activities

- **Editorial Board Member** for EPJ-ST special issue: “Radio Detection of Ultra-high Energy Neutrino & Cosmic Ray” (15 articles) with M.S. Athar, D. Besson, V. Grachev, and I. Karvchenko (2024)
- **Peer Reviewer** for Springer’s EPJ-ST journal (10 reviewed manuscripts) (2024-2025)
- **Founder and Editor-in-Chief** of *The Magnetic Monopole* e-magazine for KU’s Society of Physics Students (June 2022) [View Issue]

Professional Memberships

- **American Physical Society (APS)**: Graduate Student Member (2020-Present)
- **Physics & Astronomy Graduate Student Organization (GSO), KU**:
 - Graduate Students Representative - Member at Large (2025-Present)
 - Graduate Students Representative - Liaison (2024-2025)
 - GSO Executive Officer (2023-2024)
- **Sir Syed Global Scholar Award (SSGSA)**:
 - Team Lead, Promotions & Outreach (2023-Present)
 - Member, Mentorship & Applications Team (2021-2023)
 - AMU Campus Team Member (2019-2020)
- **Society of Physics Students**: KU Chapter Graduate President (2021-2022)
- **Sigma Pi Sigma**: Lifetime Member (National Physics Honor Society)
- **National Society of Black Physicists**: Student Member (2021-2023)
- **AMU Society of Physics**: Member (2018-Present)

Technical Proficiencies

- **Programming Languages**: C/C++, Python, FORTRAN
- **Physics Software**: CERN ROOT, AraRoot, Wolfram Mathematica, HFSS
- **Operating Systems**: Linux (Ubuntu), Windows, macOS
- **Document Preparation**: L^AT_EX, Microsoft Office Suite
- **Research Methods**: Neutrino detection and analysis techniques, radio antennas, RF engineering

Honors & Awards

Academic Honors

- **University Gold Medals:**
 - M.Sc. Physics (Top among 50 students), AMU (2020)
 - B.Sc. Honors Physics (Top among 120 students), AMU (2018)
 - Highest percentage in B.Sc. (Hons) Faculty of Science, AMU (2018)
- **Abdul Aziz Medal** for B.Sc. Physics, AMU (2018)
- **Undergraduate Topper Scholarship**, UGC India (2020)
- **University Merit Financial Award**, AMU (2019-2020)
- **First Position** in Higher Secondary (12th) among 40 students
- **Certificate of Merit** for perfect 10/10 CGPA in Secondary (10th)

Fellowships & Grants

- **Ralston Dissertation Fellowship**, KU Physics (2025-26)
- **McKeithan Summer Fellowship**, KU Physics (2025)
- **Graduate Student Award for Distinguished Service**, KU (2024-2025)
- **Wallace S Strobel Scholarship**, KU EECS (2024-2025)
- **Ralston Summer Fellowship**, KU Physics (2024)
- **Doctoral Student Research Fund**, KU (2024) for Summit station Greenland visit
- **Distinguished Students Award 2021-2022**, APS FIP
- **Carl & Catherine Chaffee Scholarship**, KU Physics (2020-2021)

Conference Travel Awards

- **Graduate Student Travel Fund**, KU Physics (2023, 2025)
- **DAP Travel Grant 2024** and **DPF Travel Award 2024**, APS
- **DPF Travel Grant 2022**, APS
- **CLAS Graduate Scholarly Development Fund 2022**, KU

Other Distinctions

- **Washington DCs Scholarship** (2015-2019) from Aligarh Alumni Association
- **Sir Syed Global Scholar Award** (2018) for prospective graduate students

Outreach & Leadership

- Co-organized India's inaugural workshop on **Radio Detection of UHE Neutrino and Cosmic Rays** at AMU (April 2024)
- Conducted outreach seminar on **Physics Research Opportunities** at AMU (2021)
- Scored 10/10 in India's first **Science Leadership Workshop** (2020)
- Volunteer for **One Person One Tree** plantation drive (2017)
- Participant in **Blood Donation Camp** (2017)

Invited Lectures & Seminars

- **Radio Engineering for Particle Astrophysics Experiments**
HEP Seminar, University of Nebraska-Lincoln (March 26, 2025)
- **Radio Engineering for Particle Astrophysics Experiments**
Astroparticle Seminar, Ohio State University (January 22, 2025)

- **Radio Engineering & Antenna Physics**
Lecture Series, IEI Summer School 2024, University of Alaska, Anchorage (July 2-5, 2024)
- **UHE Neutrino & Cosmic Ray Detectors**
ICEHAP Seminar, Chiba University, Japan (August 9, 2023)
- **Antenna Physics & Measurements**
Lecture Series, IEI Summer School 2023, South Dakota School of Mines (July 2-5, 2023)
- **UHE Neutrino & Cosmic Ray Detectors**
Physics Colloquium, AMU, India (January 12, 2023)
- **Antennas in UHE Neutrino Detectors**
Particle Physics Seminar, University of Delaware (October 4, 2022)

Conference Presentations

Oral Presentations

- **RNO-G in-ice Antennas**
RNO-G Collaboration Meeting, Uppsala University (September 9-14, 2024)
- **Radio Frequency Instrumentation**
Radio Neutrino Workshop, AMU, India (April 18-20, 2024)
- **Towards Five Station ARA Analysis**
APS April Meeting, Sacramento (April 3-6, 2024)
- **ARA Antenna Models**
ARA Collaboration Meeting, Ohio State University (March 5-8, 2024)
- **Current Status & Future of ARA**
APS Prairie Section Meeting, U Missouri (November 30-December 2, 2023)
- **Introduction to Antenna Physics**
RET Collaboration Meeting, Vrije University Brussels (November 8-11, 2023)
- **Gain Calibration of RNO-G Antennas**
RNO-G Fall Meeting, Penn State (September 11-15, 2023)
- **Galactic Noise in Radio Neutrino Experiments**
Astro Seminar, University of Kansas (October 28, 2022)
- **Towards ARA Station 1 Calibration**
APS April Meeting, New York (April 9-12, 2022)
- **Ice Calibration Planning**
RNO-G Collaboration Meeting, U Chicago (April 8, 2022)

Poster Presentations

- **Measurements of LPDA Antennas**
APS April Meeting, Sacramento (April 3-6, 2024)
- **Overview of ARA Detector**
ICTS Neutrino School, India (April 22-May 3, 2024)
- **Calibration of ARA Station 1**
38th ICRC, Nagoya University, Japan (July 29, 2023)

Professional Development

- **Responsible Conduct of Research**, (CITI Program, 2021) - 100%
- **From the Big Bang to Dark Energy** (U Tokyo Coursera, 2020) - 100%
- **Particle Physics: Introduction** (U Geneva Coursera, 2019) - 100%
- **Senior Diploma in Painting** (Bangiya Sangeet Parishad)